```
import numpy as np
import pandas as pd
```

```
df = pd.read_csv("/content/sales_dataset.csv")
```

```
df.head()
```

	Order_ID	Order_Date	Product	Category	Quantity	Unit_Price	Sales_Amount	Region
0	1001	2026-01-05	Laptop	Electronics	2	50000.0	100000	South
1	1002	2026-01-06	Smartphone	Electronics	3	20000.0	60000	North
2	1003	2026-01-07	Office Chair	Furniture	5	3000.0	15000	East
3	1004	2026-01-08	Headphones	Electronics	4	1500.0	6000	West
4	1005	2026-01-09	Desk	Furniture	2	7000.0	14000	South

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```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 8 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Order_ID      15 non-null     int64
 1   Order_Date    15 non-null     object
 2   Product       15 non-null     object
 3   Category      15 non-null     object
 4   Quantity      14 non-null     object
 5   Unit_Price    14 non-null     float64
 6   Sales_Amount  15 non-null     int64
 7   Region        15 non-null     object
dtypes: float64(1), int64(2), object(5)
memory usage: 1.1+ KB
```

```
df.isnull().sum()
```

	0
Order_ID	0
Order_Date	0
Product	0
Category	0
Quantity	1
Unit_Price	1
Sales_Amount	0
Region	0

dtype: int64

```
df
```

	Order_ID	Order_Date	Product	Category	Quantity	Unit_Price	Sales_Amount	Region
0	1001	2026-01-05	Laptop	Electronics	2	50000.0	100000	South
1	1002	2026-01-06	Smartphone	Electronics	3	20000.0	60000	North
2	1003	2026-01-07	Office Chair	Furniture	5	3000.0	15000	East
3	1004	2026-01-08	Headphones	Electronics	4	1500.0	6000	West
4	1005	2026-01-09	Desk	Furniture	2	7000.0	14000	South
5	1006	2026-01-10	Tablet	Electronics	NaN	18000.0	36000	North
6	1007	2026-01-11	Laptop	Electronics	1000	50000.0	50000000	East
7	1008	2026-01-12	Smartphone	electronics	2	20000.0	40000	south
8	1009	2026-01-13	Office Chair	Furniture	3	3000.0	9000	West
9	1009	2026-01-13	Office Chair	Furniture	3	3000.0	9000	West
10	1011	2026-01-15	Monitor	Electronics	2	NaN	24000	North
11	1012	2026-01-16	Keyboard	Electronics	5	800.0	3500	East
12	1013	2026-01-17	Printer	Electronics	-2	12000.0	-24000	West
13	1014	2026-13-40	Desk	Furniture	2	7000.0	14000	South
14	1015	2026-01-19	Mouse	Electronics	Three	500.0	1500	N0rth

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```
df.duplicated()
```

	0
0	False
1	False
2	False
3	False
4	False
5	False
6	False
7	False
8	False
9	True
10	False
11	False
12	False
13	False
14	False

dtype: bool

```
df.duplicated().sum()
```

```
np.int64(1)
```

```
df_duplicates = df[df['Order_ID'].duplicated()]
print(df_duplicates)
```

```
   Order_ID  Order_Date       Product   Category Quantity  Unit_Price  \
9      1009  2026-01-13  Office Chair  Furniture        3      3000.0

   Sales_Amount Region
9          9000   West
```

```
highest_priced_product = df[df['Unit_Price'] == df['Unit_Price'].max()]
print("Highest Priced Product:\n ",highest_priced_product[['Product','Unit_Price']])
```

```
Highest Priced Product:
    Product  Unit_Price
0  Laptop     50000.0
6  Laptop     50000.0
```

```
df['Region'] = df['Region'].str.strip().str.title()
df['Category'] = df['Category'].str.strip().str.title()

df["Region"] = df["Region"].replace({
    "N0Rth": "North",
    "N0rth": "North"
})

df[['Region','Category']]
```

	Region	Category
0	South	Electronics
1	North	Electronics
2	East	Furniture
3	West	Electronics
4	South	Furniture
5	North	Electronics
6	East	Electronics
7	South	Electronics
8	West	Furniture
9	West	Furniture
10	North	Electronics
11	East	Electronics
12	West	Electronics
13	South	Furniture
14	North	Electronics

```
duplicate_rows = df[df.duplicated()]
print(duplicate_rows)
```

```
   Order_ID  Order_Date       Product   Category Quantity  Unit_Price  \
9      1009  2026-01-13  Office Chair  Furniture        3      3000.0

   Sales_Amount Region
9          9000   West
```

```
df = df.drop_duplicates()

df.shape
```

```
(14, 8)
```

```
numeric_columns = ["Order_ID", "Quantity", "Unit_Price", "Sales_Amount"]

for col in numeric_columns:
    df[col] = pd.to_numeric(df[col], errors="coerce")

df["Order_Date"] = pd.to_datetime(
    df["Order_Date"],
    format="%Y-%m-%d",
    errors="coerce"
)

print(df.dtypes)
```

```
Order_ID                int64
Order_Date      datetime64[ns]
Product                 object
```

```
Category               object
Quantity              float64
Unit_Price            float64
Sales_Amount            int64
Region                 object
dtype: object
```

```
df
```

	Order_ID	Order_Date	Product	Category	Quantity	Unit_Price	Sales_Amount	Region
0	1001	2026-01-05	Laptop	Electronics	2.0	50000.0	100000	South
1	1002	2026-01-06	Smartphone	Electronics	3.0	20000.0	60000	North
2	1003	2026-01-07	Office Chair	Furniture	5.0	3000.0	15000	East
3	1004	2026-01-08	Headphones	Electronics	4.0	1500.0	6000	West
4	1005	2026-01-09	Desk	Furniture	2.0	7000.0	14000	South
5	1006	2026-01-10	Tablet	Electronics	NaN	18000.0	36000	North
6	1007	2026-01-11	Laptop	Electronics	1000.0	50000.0	50000000	East
7	1008	2026-01-12	Smartphone	Electronics	2.0	20000.0	40000	South
8	1009	2026-01-13	Office Chair	Furniture	3.0	3000.0	9000	West
10	1011	2026-01-15	Monitor	Electronics	2.0	NaN	24000	North
11	1012	2026-01-16	Keyboard	Electronics	5.0	800.0	3500	East
12	1013	2026-01-17	Printer	Electronics	-2.0	12000.0	-24000	West
13	1014	NaT	Desk	Furniture	2.0	7000.0	14000	South
14	1015	2026-01-19	Mouse	Electronics	NaN	500.0	1500	North

Next steps: Generate code with df | New interactive sheet

```
revenue_by_region = df.groupby('Region')['Sales_Amount'].sum()
print(revenue_by_region)
```

```
Region
East     50018500
North      121500
South      168000
West        -9000
Name: Sales_Amount, dtype: int64
```

```
df = df.dropna()
df
```

	Order_ID	Order_Date	Product	Category	Quantity	Unit_Price	Sales_Amount	Region
0	1001	2026-01-05	Laptop	Electronics	2.0	50000.0	100000	South
1	1002	2026-01-06	Smartphone	Electronics	3.0	20000.0	60000	North
2	1003	2026-01-07	Office Chair	Furniture	5.0	3000.0	15000	East
3	1004	2026-01-08	Headphones	Electronics	4.0	1500.0	6000	West
4	1005	2026-01-09	Desk	Furniture	2.0	7000.0	14000	South
6	1007	2026-01-11	Laptop	Electronics	1000.0	50000.0	50000000	East
7	1008	2026-01-12	Smartphone	Electronics	2.0	20000.0	40000	South
8	1009	2026-01-13	Office Chair	Furniture	3.0	3000.0	9000	West
11	1012	2026-01-16	Keyboard	Electronics	5.0	800.0	3500	East
12	1013	2026-01-17	Printer	Electronics	-2.0	12000.0	-24000	West

Next steps: Generate code with df | New interactive sheet

```
negative_sales = df[df["Quantity"] < 0]

print(negative_sales[
    ["Order_ID", "Product", "Quantity",
```

```
     "Unit_Price", "Sales_Amount"]
])
```

```
    Order_ID  Product  Quantity  Unit_Price  Sales_Amount
12      1013  Printer      -2.0     12000.0        -24000
```

```
df["Quantity_Flag"] = np.where(
    df["Quantity"] < 0,
    "Not Valid",
    "Valid"
)

print(df[["Order_ID", "Quantity", "Quantity_Flag"]])
```

```
    Order_ID  Quantity Quantity_Flag
0       1001       2.0         Valid
1       1002       3.0         Valid
2       1003       5.0         Valid
3       1004       4.0         Valid
4       1005       2.0         Valid
6       1007    1000.0         Valid
7       1008       2.0         Valid
8       1009       3.0         Valid
11      1012       5.0         Valid
12      1013      -2.0     Not Valid
/tmp/ipykernel_2151/3669730383.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-ver
  df["Quantity_Flag"] = np.where(
```

```
df
```

	Order_ID	Order_Date	Product	Category	Quantity	Unit_Price	Sales_Amount	Region	Quantity_Flag
0	1001	2026-01-05	Laptop	Electronics	2.0	50000.0	100000	South	Valid
1	1002	2026-01-06	Smartphone	Electronics	3.0	20000.0	60000	North	Valid
2	1003	2026-01-07	Office Chair	Furniture	5.0	3000.0	15000	East	Valid
3	1004	2026-01-08	Headphones	Electronics	4.0	1500.0	6000	West	Valid
4	1005	2026-01-09	Desk	Furniture	2.0	7000.0	14000	South	Valid
6	1007	2026-01-11	Laptop	Electronics	1000.0	50000.0	50000000	East	Valid
7	1008	2026-01-12	Smartphone	Electronics	2.0	20000.0	40000	South	Valid
8	1009	2026-01-13	Office Chair	Furniture	3.0	3000.0	9000	West	Valid
11	1012	2026-01-16	Keyboard	Electronics	5.0	800.0	3500	East	Valid
12	1013	2026-01-17	Printer	Electronics	-2.0	12000.0	-24000	West	Not Valid

Next steps: Generate code with df | New interactive sheet

```
revenue_by_region = df[df['Quantity_Flag']=="Valid"].groupby("Region")['Sales_Amount'].sum()
print(revenue_by_region)
```

```
Region
East     50018500
North       60000
South      154000
West        15000
Name: Sales_Amount, dtype: int64
```

```
valid_df = df[df["Quantity_Flag"] == "Valid"].copy()

Q1 = valid_df["Unit_Price"].quantile(0.25)
Q3 = valid_df["Unit_Price"].quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = valid_df[
```

```
    (valid_df["Unit_Price"] < lower_bound) |
    (valid_df["Unit_Price"] > upper_bound)
]

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower bound:", lower_bound)
print("Upper bound:", upper_bound)

print("\nOutliers:")
print(outliers[["Order_ID", "Product", "Quantity","Unit_Price"]])
```

```
Q1: 3000.0
Q3: 20000.0
IQR: 17000.0
Lower bound: -22500.0
Upper bound: 45500.0

Outliers:
   Order_ID Product  Quantity  Unit_Price
0      1001  Laptop       2.0     50000.0
6      1007  Laptop    1000.0     50000.0
```

```
Q1 = valid_df["Sales_Amount"].quantile(0.25)
Q3 = valid_df["Sales_Amount"].quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = valid_df[
    (valid_df["Sales_Amount"] < lower_bound) |
    (valid_df["Sales_Amount"] > upper_bound)
]

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower bound:", lower_bound)
print("Upper bound:", upper_bound)

print("\nOutliers:")
print(outliers[["Order_ID", "Product", "Quantity","Unit_Price","Sales_Amount"]])
```

```
Q1: 9000.0
Q3: 60000.0
IQR: 51000.0
Lower bound: -67500.0
Upper bound: 136500.0

Outliers:
   Order_ID Product  Quantity  Unit_Price  Sales_Amount
6      1007  Laptop    1000.0     50000.0      50000000
```

```
valid_df.to_csv("valid_sales_dataset.csv",index=False)
```

```
valid_df
```

	Order_ID	Order_Date	Product	Category	Quantity	Unit_Price	Sales_Amount	Region	Quantity_Flag
0	1001	2026-01-05	Laptop	Electronics	2.0	50000.0	100000	South	Valid
1	1002	2026-01-06	Smartphone	Electronics	3.0	20000.0	60000	North	Valid
2	1003	2026-01-07	Office Chair	Furniture	5.0	3000.0	15000	East	Valid
3	1004	2026-01-08	Headphones	Electronics	4.0	1500.0	6000	West	Valid
4	1005	2026-01-09	Desk	Furniture	2.0	7000.0	14000	South	Valid
6	1007	2026-01-11	Laptop	Electronics	1000.0	50000.0	50000000	East	Valid
7	1008	2026-01-12	Smartphone	Electronics	2.0	20000.0	40000	South	Valid
8	1009	2026-01-13	Office Chair	Furniture	3.0	3000.0	9000	West	Valid
11	1012	2026-01-16	Keyboard	Electronics	5.0	800.0	3500	East	Valid